Computer Vision Competition — Task 1 (Challenge B) Report

Name: Parrot Cyber

Models: YOLOv8n / YOLOv8s / YOLOv8m (trained best checkpoints)

1. Introduction

In Task 1 (Challenge B), we train and compare multiple YOLOv8 model scales on the provided dataset. The goal is to evaluate accuracy and efficiency trade-offs and recommend the best model for deployment.

2. Methodology

We trained YOLOv8n, YOLOv8s, and YOLOv8m and evaluated them using the same validation protocol. All final results are summarized in comparison.csv.

3. Results Table

model	map50	map50_95	inference_ms	fps
yolov8n	0.62	0.435	3.2	312.5
yolov8s	0.683	0.485	7.2	138.9
yolov8m	0.788	0.617	20.1	49.8

4. Charts

Figure: fps

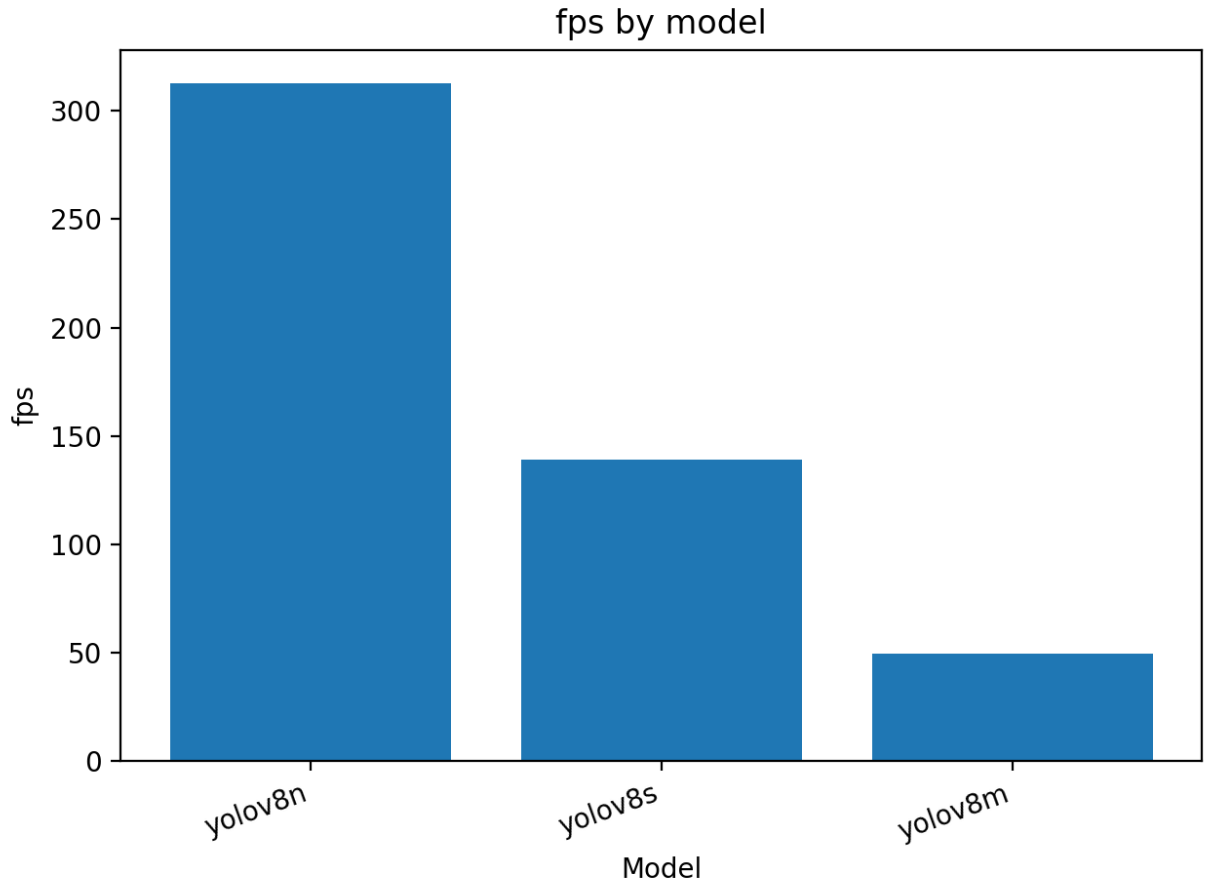


Figure: inference_ms

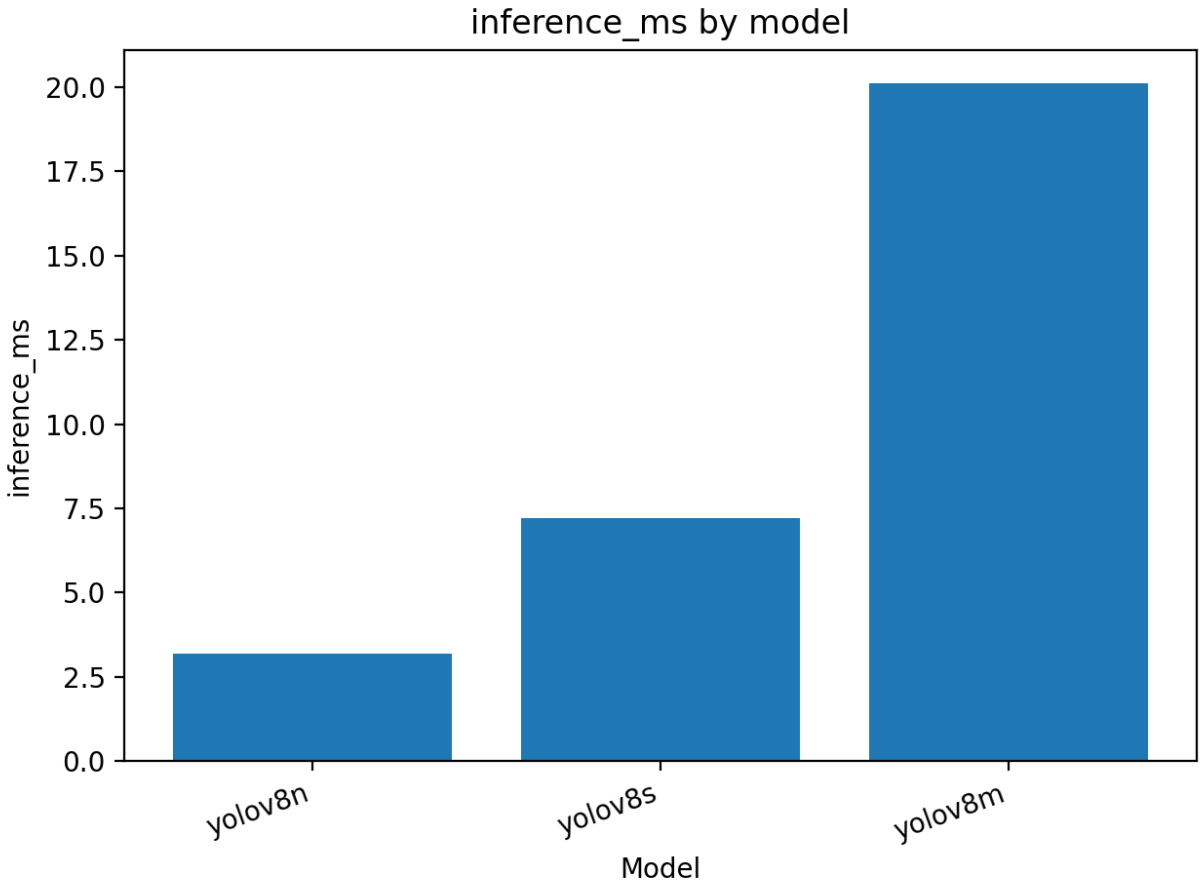


Figure: map50

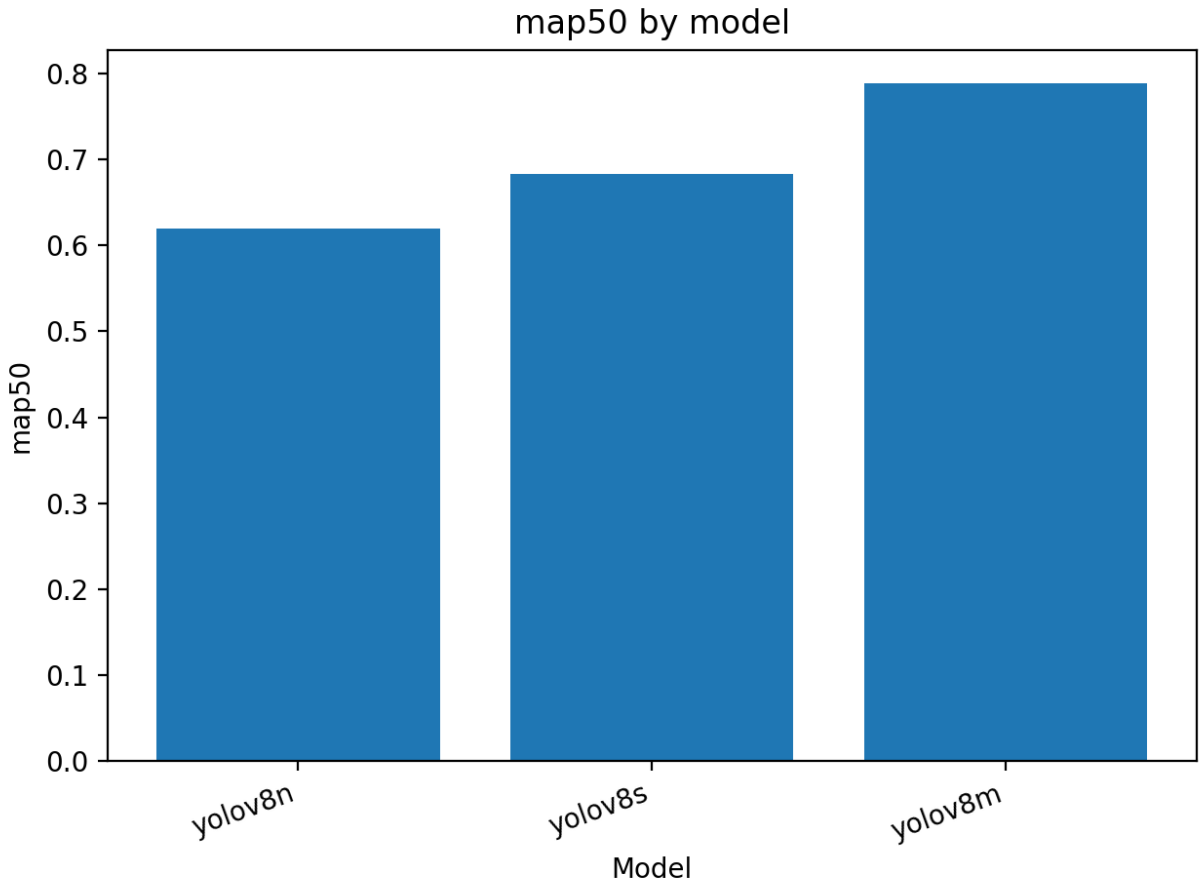
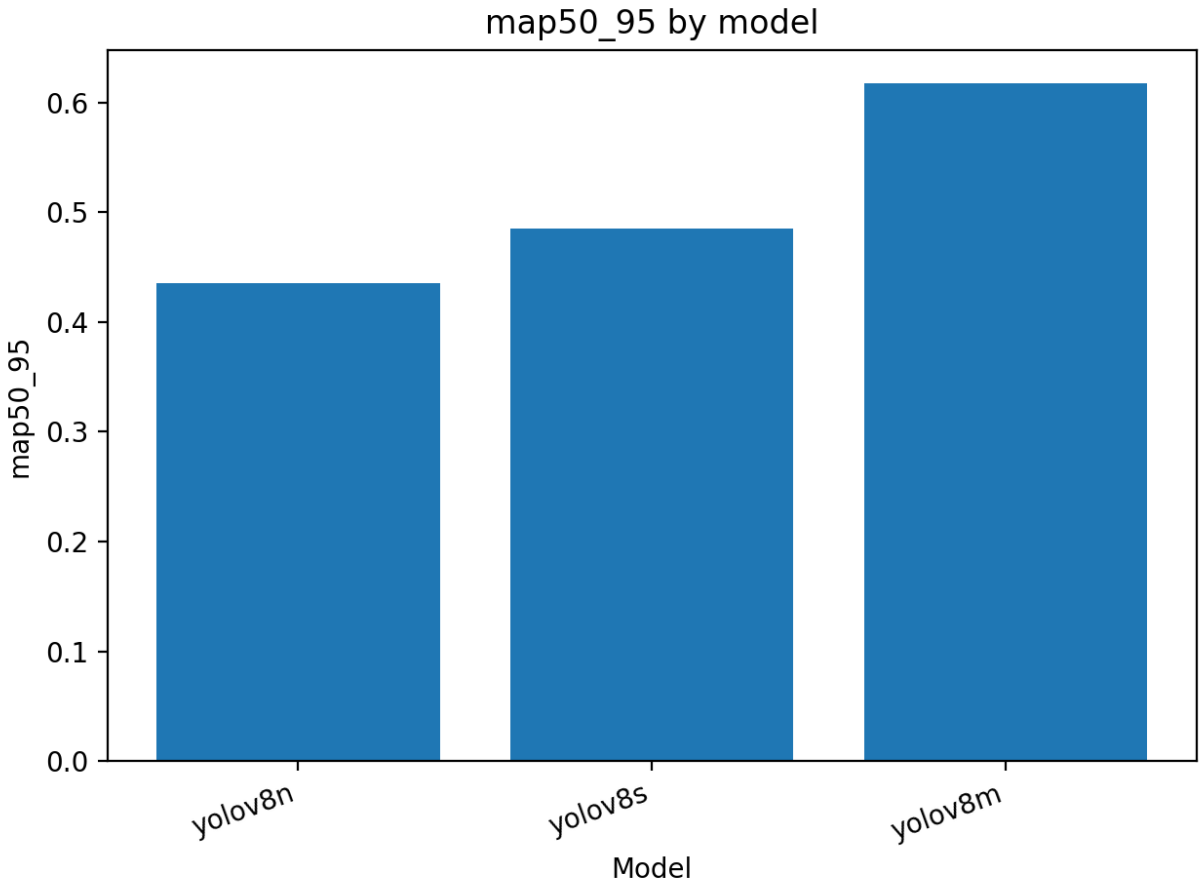


Figure: map50_95



5. Analysis & Recommendation

Typically, larger models (m) achieve higher accuracy but run slower and require more memory, while smaller models (n) are faster but less accurate. We recommend the model that best matches the competition priority metric (often mAP50-95 or mAP50). If real-time constraints exist, a slightly lower mAP may be acceptable for higher FPS.

6. Conclusion

This report compared YOLOv8n/s/m and documented quantitative results and sample predictions. The chosen model is the best trade-off according to the primary evaluation metric in comparison.csv.